# AI – DRIVEN SMART CINNAMON GROWTH MONITORING & AUTOMATED HARVESTING SYSTEM

## INDIVIDUAL THESIS

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Individual component: AI – Guided Robotic Machine for Automated Cinnamon Bark Harvesting

Group ID: R26-IT-102

Report submitted in partial fulfilment of the requirements for the

Bachelor of Science (Hons) in Information Technology

Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

March 2026

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# ABSTRACT

Cinnamon cultivation in Sri Lanka is a globally recognised agricultural industry, but bark harvesting still depends on manual labour and skilled workers. Identifying suitable bark regions and peeling them precisely takes expertise and time, and results in inconsistent quality, injury risk and rising labour cost. This thesis presents the design, implementation and evaluation of an AI-guided robotic machine for automated cinnamon bark harvesting, developed as the individual component of the group project R26-IT-102.

A camera mounted above the work plane captures images of cinnamon stems. After pre-processing with OpenCV, a CNN-based detector, initialised from pre-trained weights and compared with a small baseline CNN, identifies peelable bark regions, and classical edge and contour analysis extracts a cut line within the selected region. Camera calibration and a plane homography convert the cut line into robot coordinates, and inverse kinematics drives a five-degree-of-freedom arm with MG996R servos under layered safety controls. A harvest gate uses the readiness output of the project's prediction models, and outcomes are reported to the cloud dashboard. The system was evaluated at model, geometry and system levels using a self-collected dataset of 2,000 images and bench trials on ⟨N⟩ stems.

On the held-out test set of 300 images, the detector achieved a precision of ⟨P⟩ and a recall of ⟨R⟩, compared with ⟨P_b⟩ and ⟨R_b⟩ for the baseline CNN. The mean end-effector positioning error was ⟨E⟩ mm, with ⟨W⟩% of trials within 5 mm. Of ⟨N⟩ bench trials on cut stems, ⟨k⟩ ended in successful bark extraction (⟨S⟩%, 95% CI ⟨L⟩–⟨U⟩%), with a mean cycle time of ⟨T⟩ s. The work demonstrates that a low-cost, camera-guided five-degree-of-freedom arm can locate and slit cinnamon bark on cut stems under controlled conditions. Its main limitations are the limited accuracy of hobby servo motors, sensitivity to lighting and a small single-species dataset. Future work should add depth sensing, force feedback, larger multi-site datasets and field trials.

**Keywords:** Artificial intelligence, computer vision, robotic harvesting, cinnamon bark, agricultural automation, deep learning

# ACKNOWLEDGEMENTS

The successful completion of this research would not have been possible without the support and guidance of several individuals and organizations. First and foremost, I would like to express my sincere gratitude to my research supervisor and academic advisors for their continuous guidance, encouragement, and valuable feedback throughout the research process. Their expertise and insights have played a significant role in shaping the direction and quality of this research. I would also like to thank the faculty members of the Information Technology program for providing the academic knowledge and research environment necessary to conduct this study. Their support and resources have greatly contributed to the successful development of this project. My appreciation is extended to my research team members for their collaboration and dedication during the project. The teamwork and knowledge sharing among members have helped in integrating the IoT, machine learning, and artificial intelligence components of the system effectively. Finally, I would like to express my heartfelt gratitude to my family and friends for their constant encouragement, patience, and moral support throughout the completion of this research project.

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# LIST OF ABBREVIATIONS

| Abbreviation | Meaning |
|---|---|
| AI | Artificial Intelligence |
| API | Application Programming Interface |
| CNN | Convolutional Neural Network |
| DOF | Degrees of Freedom |
| HSV | Hue, Saturation, Value (colour space) |
| HTTPS | Hypertext Transfer Protocol Secure |
| IK | Inverse Kinematics |
| IoT | Internet of Things |
| IoU | Intersection over Union |
| JSON | JavaScript Object Notation |
| LSTM | Long Short-Term Memory |
| mAP | Mean Average Precision |
| ML | Machine Learning |
| NDVI | Normalized Difference Vegetation Index |
| NMS | Non-Maximum Suppression |
| PCA | Principal Component Analysis |
| PWM | Pulse-Width Modulation |
| REST | Representational State Transfer |
| RGB | Red Green Blue (Image Channels) |
| ROS | Robot Operating System |
| SDG | Sustainable Development Goal |
| UV | Ultraviolet Light |
| YOLO | You Only Look Once (object detector family) |

# 1 INTRODUCTION

## 1.1 Overall project context

Sri Lanka is the traditional centre of "true" cinnamon (*Cinnamomum verum*) production, and cinnamon bark is one of the country's most recognisable agricultural exports [CITE: Export Development Board / Department of Export Agriculture figures on production and export value]. Despite this importance, cultivation and harvesting still depend largely on manual practice and inherited skill. Growth conditions are rarely measured, disease is usually noticed late, the decision about when a stem is ready to be peeled is made by eye and touch, and the peeling itself is slow, skill-dependent and physically demanding. The consequences are variable yield, inconsistent bark quality and a shrinking pool of skilled peelers.

The group project R26-IT-102, *AI-Driven Multi-Environment Cinnamon Growth Monitoring and Automated Harvesting System Using Robotic Manipulation and Multispectral Sensing*, addresses this problem as one integrated platform rather than as a set of isolated tools. The platform connects (i) Internet of Things (IoT) controlled growth environments that record temperature, humidity, soil moisture and light, (ii) machine-learning models that forecast growth and harvest readiness and detect disease, (iii) an AI-guided robotic machine that locates and extracts cinnamon bark, and (iv) a cloud dashboard through which farmers and technicians monitor the whole workflow. The work is divided among three members. Member 1 builds the IoT environment and its sensing and cloud layer. Member 2 builds the machine-learning models for growth prediction, harvest readiness and disease detection. Member 3, the author of this thesis, builds the AI vision and robotic harvesting component. Figure 1.1 shows how the three components relate.

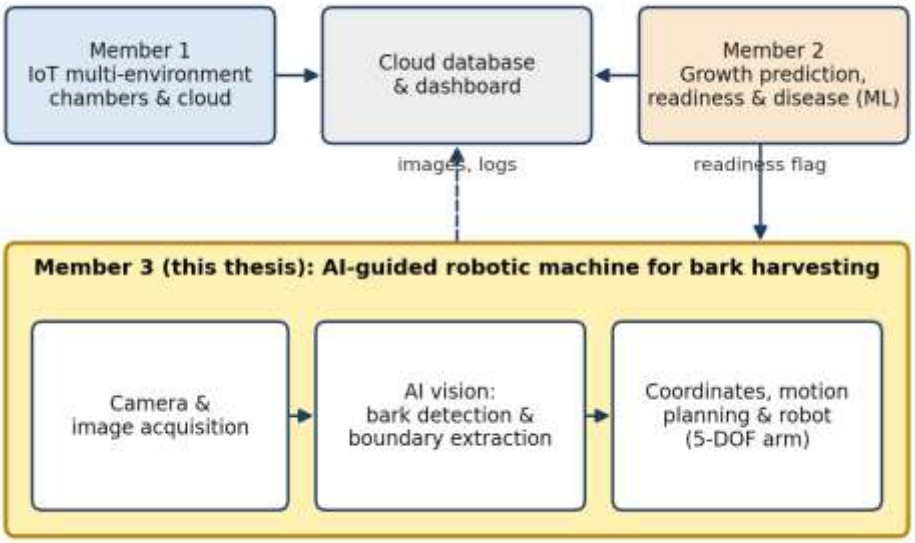



**Figure 1.1: Overall project architecture and the scope of the Member 3 component (highlighted)**

### 1.2 Individual component context

The individual component reported in this thesis is the **AI-guided robotic machine for automated cinnamon bark harvesting**. Its purpose is to take over the part of the harvesting chain that currently depends most on human skill: finding the region of a cinnamon stem whose bark can be extracted and then guiding a mechanical arm to extract it in a controlled way. The component receives images from a camera mounted on the robotic machine and, where available, a harvest-readiness indication from Member 2's models. It produces a detected bark region, a cut line expressed in the robot's coordinate frame, a sequence of joint commands for the arm, and a record of what was done so that the result can be shown on the dashboard.

In traditional practice the work proceeds roughly as follows: mature stems are selected and cut, leaves and side shoots are removed, the outer layer is scraped off, the stem is rubbed to loosen the bark, the bark is slit lengthwise and peeled off, and the pieces are later dried into quills [VALIDATE with field-visit notes and cite a primary source]. The prototype in this thesis concentrates on the identification, slitting and peeling steps on a bench-scale rig, because these are the steps in which a vision system can replace a trained eye and a controlled mechanism can replace repetitive manual effort.

### 1.3 Problem statement

Cinnamon bark extraction is performed manually by skilled workers. Judging which part of a stem is suitable for peeling, and carrying out the peel without tearing the bark, depends on experience and is therefore slow, inconsistent between workers and physically hard on the body, with a risk of blade injuries and repetitive-strain problems. Skilled peelers are becoming harder to find and more expensive during the harvest season. Existing agricultural robots and vision systems are designed mainly for fruit picking, weeding or crop monitoring; they locate a discrete object and detach it, whereas bark extraction requires the boundary of a continuous surface to be found and a cut and peel to be carried out along it. As a result, no ready-made solution exists that can detect cinnamon bark regions from images and translate that detection into controlled robotic extraction.

The problem addressed by this research is therefore: *how can an affordable, camera-based AI system detect the peelable bark region of a cinnamon stem and generate accurate, safe motion commands so that a low-cost robotic arm can extract the bark consistently and with less manual effort?*

### 1.4 Research gap

The literature and the available products reviewed in Chapter 2 leave four gaps that this component is designed to address.

- **Bark-based harvesting is largely unexplored.** Agricultural harvesting research concentrates on fruit and vegetables. Very little work addresses bark extraction, which needs precise detection of bark boundaries and a controlled peeling action.
- **Rule-based image processing is fragile.** Many harvesting systems rely on colour thresholds or hand-written rules that degrade when lighting, growth stage or bark texture changes. A learned detector, supported by classical boundary extraction, has not been evaluated for cinnamon.
- **Vision, prediction and control are seldom integrated.** Most studies treat detection, forecasting and robot control separately. A platform in which a readiness prediction gates a vision-guided robot has not been demonstrated for cinnamon.

- **Affordability.** Commercial multispectral cameras and industrial robot arms are expensive. A solution built from commodity cameras, servo motors and open-source software is needed if it is ever to reach smallholders.

### 1.5 Aim and objectives

**Aim.** To design, implement and evaluate an AI vision-guided robotic harvesting system that detects peelable cinnamon bark regions and performs automated bark extraction with high precision and efficiency on a bench-scale prototype.

**Specific objectives.** The objectives of the proposal were refined into measurable form as follows.

**RO1** Develop an AI-based image-processing pipeline that detects cinnamon bark regions and extracts their boundaries from plant-stem images.

**RO2** Train and evaluate a deep-learning (CNN-based) model that identifies the optimal bark region for harvesting, and compare it with a baseline CNN.

**RO3** Convert image-space detections into accurate harvesting coordinates in the robot's workspace through camera calibration.

**RO4** Integrate the vision module with the control system of a five-degree-of-freedom robotic arm and with the readiness output of Member 2's models.

**RO5** Evaluate the integrated system using detection, positioning, peeling-quality and efficiency metrics, including a comparison with manual peeling where feasible.

### 1.6 Research questions or hypotheses

**RQ1** How accurately can a CNN-based detector, supported by classical boundary extraction, identify peelable bark regions on cinnamon stems under varying lighting and background conditions?

**RQ2** How accurately can image-space detections be converted into robot-space coordinates for a low-cost servo-driven arm using a single calibrated camera?

**RQ3** Can the integrated pipeline (readiness gate, detection, coordinate generation and peeling) complete bark extraction on the bench prototype with an acceptable success rate and cycle time?

The associated working hypotheses, to be accepted or rejected on the evidence in Chapter 6, are:

**H1** A transfer-learning detector achieves higher image-level precision and recall on held-out stems than the baseline CNN of Table 4.5.

**H2** Calibrated coordinate mapping places the end-effector within the positioning tolerance defined in Table 3.2 in at least 90% of trials.

**H3** The integrated system completes bark extraction without operator intervention in at least the success rate defined in Table 3.2.

### 1.7 Scope limitations and assumptions

**In scope.** Image acquisition and pre-processing; training and evaluation of the bark-detection model; boundary extraction; camera calibration and pixel-to-robot coordinate mapping; inverse kinematics and motion control of a five-degree-of-freedom arm; a peeling end-effector; integration with the IoT/cloud layer and with the readiness output of Member 2; bench-scale testing.

**Out of scope.** Design of the IoT growth chambers and cloud platform (Member 1); growth-forecasting, readiness and disease-detection models (Member 2); autonomous navigation in a plantation; cutting and transporting stems from the field; drying and quill-making; certification of the machine for commercial operation.

**Assumptions and limitations.**

- Peeling is demonstrated on cut cinnamon stems held in a fixture within the arm's workspace, not on standing plants.
- Only *Cinnamomum verum* is represented in the dataset, with images captured at cinnamon estates in the Galle and Matara districts and at the project laboratory.
- A single monocular camera is used; depth is obtained from the known fixture plane rather than from a depth sensor.

- Hobby-grade MG996R servo motors limit stiffness and repeatability; the prototype demonstrates feasibility, not industrial throughput.
- The readiness signal from Member 2 is treated as an external input, and its accuracy is not evaluated in this thesis.

### 1.8 Individual contribution

The author's contribution follows the work breakdown agreed in the project proposal (tasks M3-T1 to M3-T7):

- **M3-T1** captured and organised the cinnamon stem and bark image dataset;
- **M3-T2** developed the OpenCV pre-processing module (noise removal, lighting normalisation, resizing, augmentation);
- **M3-T3** built, trained and tuned the CNN-based bark-detection model and its baseline;
- **M3-T4** implemented identification of the optimal harvesting region and boundary extraction;
- **M3-T5** implemented camera calibration and generation of harvesting coordinates for the robot;
- **M3-T6** integrated the AI module with the robot controller and with the cloud/dashboard interfaces; and
- **M3-T7** designed and executed the bench tests of the automated harvesting process.

Mechanical construction and sensor wiring were carried out by the author. Interfaces to Member 1 and Member 2 were agreed jointly, and the work of those members is acknowledged but not claimed here.

### 1.9 Thesis structure

Chapter 2 reviews the literature and existing systems and synthesises the research gap. Chapter 3 describes the research methodology: design, data, procedure, metrics, analysis, ethics and reproducibility. Chapter 4 presents the requirements, architecture and detailed design of the solution. Chapter 5 describes implementation, integration and testing. Chapter 6 reports the results and validation. Chapter 7 discusses the

findings, contributions and limitations. Chapter 8 covers commercialisation and sustainability. Chapter 9 concludes the thesis and outlines future directions.

# 2 LITERATURE REVIEW AND RELATED WORK

## 2.1 Review method and scope

The review covered four themes: (i) deep learning and computer vision for crop monitoring and harvesting, (ii) agricultural robotic manipulation, (iii) vision-to-robot coordinate mapping and low-cost arm control, and (iv) cinnamon cultivation and processing practice. Sources were searched in IEEE Xplore, ScienceDirect, MDPI, Google Scholar and the publications of the Department of Export Agriculture, using combinations of the terms *cinnamon*, *bark*, *peeling*, *harvesting robot*, *object detection*, *YOLO*, *hand-eye calibration* and *precision agriculture*. Priority was given to journal and conference papers published from 2015 onward, with older foundational works retained where they define a method used in this thesis. In total, ⟨n1⟩ papers were screened by title and abstract and ⟨n2⟩ were retained after full-text reading. The last search was carried out in 2026.

Papers were included if they described a method or system that could inform detection of plant parts, robotic harvesting, or low-cost manipulation, and excluded if they provided no methodological detail or evaluation. Because very few sources address cinnamon bark directly, the review necessarily draws on analogous tasks and states clearly where evidence is transferred from another crop.

## 2.2 Domain and theoretical foundations

**Cinnamon harvesting practice.** True cinnamon is harvested from coppiced shoots that are allowed to mature before cutting. A stem is considered ready when its bark separates cleanly from the wood; experienced peelers judge this from stem colour, leaf appearance and the feel of the bark during rubbing [CITE and validate with estate visits]. The difficulty for automation is that the readiness cue and the usable bark surface are both visual and subtle, and they vary between stems.

**Convolutional neural networks and object detection.** CNNs learn hierarchical visual features directly from labelled images and form the basis of modern image classification and detection [1]. One-stage detectors such as YOLO predict bounding boxes and class scores in a single pass, which suits real-time use on modest hardware [2]. Training from weights pre-trained on large datasets and fine-tuning on a small

domain dataset (transfer learning) reduces the amount of labelled data needed [3]. Lightweight backbones such as MobileNet are designed for limited compute [4]. Detection quality is expressed through intersection-over-union (IoU), precision, recall and mean average precision [5].

**Classical image processing.** Colour-space conversion (for example HSV), smoothing, morphological operations and edge detection remain useful for extracting precise boundaries once a region of interest is known [6], [7]. The Canny detector is a standard choice for thin, well-localised edges [8].

**Camera calibration and plane-to-plane mapping.** Intrinsic parameters and lens distortion are estimated from images of a planar pattern [9]. When the target lies on a known plane, a planar homography maps pixel coordinates to metric coordinates on that plane [10], which removes the need for a depth sensor in a fixture-based set-up.

**Manipulator kinematics and servo control.** Forward kinematics gives the end-effector pose from joint angles, and inverse kinematics (IK) solves the reverse problem; for low-degree-of-freedom arms, closed-form or numerical IK with joint-limit checks is standard [11], [12]. Hobby servo motors are position-controlled by pulse-width modulation (PWM) and have limited torque, resolution and repeatability, which constrains achievable accuracy.

### 2.3 Related research

**Deep learning in agriculture.** Surveys by Kamilaris and Prenafeta-Boldú [13] and Liakos et al. [14] show that CNN-based methods now dominate image-based tasks in agriculture, including disease recognition, fruit counting and yield estimation. Mohanty et al. reported high accuracy for plant-disease classification from leaf images, but also noted that models trained on controlled images generalise poorly to field conditions [15]. This caution is relevant here because bark images vary strongly with lighting and background.

**Detection of harvestable targets.** Sa et al. adapted a deep detector to locate fruit for robotic picking [16], and Bargoti and Underwood applied deep detection to orchard fruit and showed the value of data augmentation and transfer learning for small datasets [17]. Both treat the target as a compact object with a clear outline. Bark is a

continuous surface whose usable region has to be delimited, which motivates the combination of a learned detector with boundary extraction adopted in this thesis.

**Harvesting robots.** Silwal et al. described a robotic apple harvester and its field evaluation [18], and Xiong et al. developed and field-tested an autonomous strawberry-harvesting robot [19]. These systems integrate perception, motion planning and a task-specific end-effector, and they report that end-effector design and perception errors in cluttered scenes are leading causes of failure. Reviews of agricultural robotics [20], [21] identify cost, robustness to variable conditions and task-specific tooling as the main barriers to adoption.

**Bark-related tasks.** Robotic work on bark-handling tasks such as rubber tapping and tree debarking exists, but these address trees in plantation or forestry settings and use different tools and materials. Searches of IEEE Xplore, ScienceDirect and Google Scholar using the terms in Section 2.1 did not locate any study on robotic peeling of cinnamon bark. Related robotic work on bark-handling tasks, such as rubber tapping and tree debarking, addresses different species, tools and settings. To the author's knowledge, no published study demonstrates vision-guided robotic peeling of cinnamon bark.

### 2.4 Existing systems products and techniques

Table 2.1 summarises the categories of existing solutions and why they do not meet the needs of cinnamon bark harvesting.

**Table 2.1: Existing solutions compared with the needs of cinnamon bark harvesting**

| Category | Typical capability | Limitation for cinnamon bark harvesting |
|---|---|---|
| Fruit-harvesting robots (apple, strawberry, tomato) | Detect discrete fruit, plan an approach, detach by cutting or gripping [18], [19] | Target is a compact object; no bark boundary detection or peeling mechanism |
| Commercial multispectral cameras (e.g. Parrot Sequoia, MicaSense RedEdge) | Vegetation indices such as NDVI for crop stress | High cost; monitor plant health but do not guide a peeling tool |
| IoT smart-farming platforms | Sensor monitoring and irrigation control | Designed for paddy, vegetables or greenhouse crops; no harvest automation |

| Category | Typical capability | Limitation for cinnamon bark harvesting |
|---|---|---|
| Sri Lankan advisory apps (e.g. Govi Nomu, Haritha) | Agronomic advice and information services [CITE] | No real-time sensing, disease prediction or harvest guidance |
| Manual and hand-tool peeling | Skilled workers peel stems with knives and rubbing tools | Slow, variable quality, injury risk, dependent on scarce skills |

### 2.5 Gap synthesis and conceptual framework

The review shows that the building blocks for this work exist: accurate CNN detectors, calibration methods that allow a single camera to guide a robot on a plane, and low-cost servo arms. What is missing is their combination for a bark-extraction task, evaluated on cinnamon. Table 2.2 links each gap to the design response in this thesis.

**Table 2.2: Research gaps and the corresponding design response**

| Gap | Design response | Where addressed |
|---|---|---|
| No bark-extraction automation | Vision-guided arm with slitting/peeling end-effector | Sections 4.4, 5.2 |
| Fragile rule-based vision | CNN detector with classical boundary refinement, trained on varied lighting | Sections 3.2, 4.4 |
| Poor integration of vision, prediction and control | Harvest gate driven by Member 2 readiness; REST interfaces to cloud | Sections 4.2, 5.3 |
| High cost | Commodity camera, MG996R servos, Arduino/Raspberry Pi, open-source software | Sections 4.3, 8.3 |

The conceptual framework is a closed loop: *perceive* (image and readiness signal), *decide* (is the stem ready and where is the usable bark), *act* (convert the cut line to joint commands and peel) and *verify* (inspect the result and log it). Each stage has a measurable output, which is what allows the evaluation in Chapter 6.

# 3 RESEARCH METHODOLOGY

## 3.1 Research design

The study follows a **design science and experimental** approach. Design science is used to build a new artefact, the AI-guided harvesting machine, through iterative design, implementation and evaluation. Experimental methods are used to measure how well the artefact performs: offline experiments for the vision model, calibration experiments for coordinate accuracy, and controlled bench trials for the integrated system. Figure 3.1 summarises the research process.

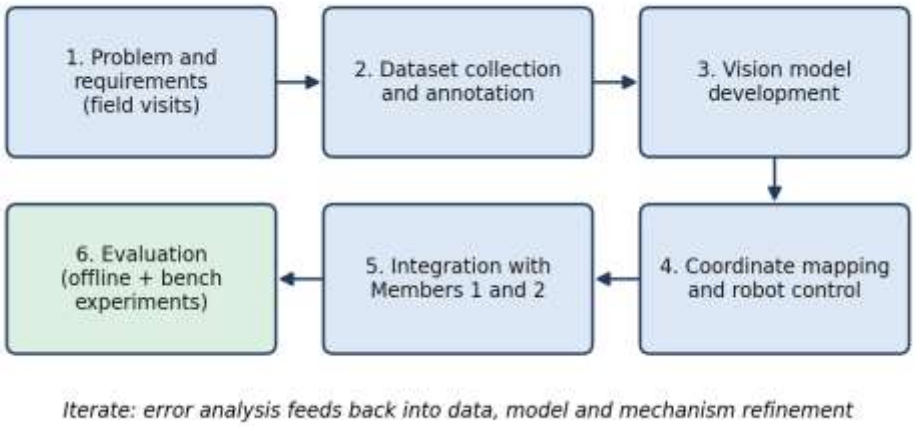



**Figure 3.1: Research process for the Member 3 component**

Three levels of evaluation are used so that failures can be traced to their source: (1) *model level*, using held-out images; (2) *geometry level*, measuring the difference between commanded and achieved end-effector positions; and (3) *system level*, measuring the outcome of complete harvest cycles. The research questions RQ1 to RQ3 correspond to these levels.

## 3.2 Data participants or study materials

No human participants are involved in the experiments. The study materials are cinnamon stems, a camera and calibration target, the robotic arm and fixture, and the image dataset described below.

**Dataset.** The dataset was planned and collected as summarised in Table 3.1. All images were captured by the project team with the project camera; public datasets were

not used for training or testing. Pre-trained weights from a public image corpus were used only to initialise the network (transfer learning).

**Table 3.1: Dataset specification for bark detection**

| Item | Specification |
|---|---|
| Image type | RGB images of cinnamon stems and bark, 1080p capture resized to 512 × 512 pixels |
| Total images | 2,000 |
| Split | Training 1,400 (70%); validation 300 (15%); test 300 (15%). Split by stem and capture session so that images of one stem never appear in more than one subset |
| Classes | peelable bark region; non-peelable region (immature, damaged or covered bark) |
| Annotation | Manual bounding boxes using LabelImg or Roboflow, following written labelling guidelines (Appendix A); a sample of 10% re-labelled by a second annotator to estimate agreement |
| Capture conditions | Natural daylight (sunny, overcast, shade), indoor white light; plain and cluttered backgrounds; several stem diameters and growth stages; multiple viewing angles |
| Augmentation | Rotation, horizontal flip and brightness adjustment, applied to the training subset only |
| Sites and dates | [Estates/locations and capture dates] |

Splitting by stem and session, and augmenting only after the split, prevents near-duplicate images from leaking from training into testing, which would inflate the measured accuracy.

**Context data.** Environmental readings (temperature, humidity, soil moisture, light intensity, timestamp) are collected by Member 1's chambers and used by Member 2's models. The Member 3 component consumes only the resulting harvest-readiness indication and does not process raw sensor data.

**Hardware materials.** A five-degree-of-freedom arm driven by MG996R servo motors, a 1080p camera mounted at a fixed pose above the work plane, a stem-holding fixture, a slitting/peeling end-effector, an Arduino or Raspberry Pi controller with servo driver, and a regulated power supply. Specifications are given in Section 4.4.

### 3.3 Experimental or study procedure

Five experiments were defined.

**E1** *Model comparison (RQ1).* Train the baseline CNN of Table 4.5 and the transfer-learning detector on the training set, tune on the validation set, and report results once on the untouched test set. Both models are compared at image level (does the image contain a peelable region?); for the detector, an image is positive if at least one peelable box exceeds the confidence threshold. Box-level metrics are reported for the detector only.

**E2** *Lighting and background robustness (RQ1).* Report test-set metrics separately for each capture condition (sunny, overcast, shade, indoor, plain and cluttered background).

**E3** *Calibration and positioning accuracy (RQ2).* Calibrate the camera; command the end-effector to [N] target points marked on the work plane; measure the offset between target and achieved position with a ruler or printed grid; repeat each point [k] times to estimate repeatability.

**E4** *Integrated bench trials (RQ3).* For [N] stems, run the complete cycle (scan, verify, plan, approach, peel, inspect) and record success, failure mode, cycle time and operator interventions.

**E5** *Latency profiling (RQ3).* Measure the time taken by each pipeline stage over [N] runs.

Where feasible, the same stems are also peeled manually by an experienced worker so that cycle time and bark integrity can be compared. [State whether this comparison was carried out and how consent and safety were handled.]

### 3.4 Measures and evaluation metrics

Detection metrics follow standard definitions [5]. With TP, FP and FN denoting true positives, false positives and false negatives at an IoU threshold of 0.5:

$$\text{Precision} = TP / (TP + FP) \qquad \text{Recall} = TP / (TP + FN) \qquad F1 = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall})$$

$$\text{IoU} = |A \cap B| / |A \cup B| \qquad \text{Positioning error} = \|p_achieved - p_target\|$$

Table 3.2 lists all measures and the acceptance targets. The targets were set by the author on the basis of the proposal and the tolerance needed to keep a blade within the

bark region; they are to be confirmed with the supervisors. [Adjust targets if the supervisors specify otherwise.]

**Table 3.2: Evaluation metrics and acceptance targets**

| Level | Metric | Definition | Target |
|---|---|---|---|
| Model | Precision, recall, F1 | At IoU ≥ 0.5 on the test set | P ≥ 0.90, R ≥ 0.85 |
| Model | mAP@0.5 | Mean average precision over classes | ≥ 0.85 |
| Model | Boundary IoU | IoU of extracted bark region against annotation | ≥ 0.80 |
| Geometry | Positioning error | Distance between target and achieved end-effector position | ≤ 5 mm in ≥ 90% of trials |
| Geometry | Repeatability | Standard deviation of repeated moves to one point | ≤ 2 mm |
| System | Peel success rate | Fraction of trials in which the bark is extracted in one piece without operator help | ≥ 80% |
| System | Bark integrity | Length and breakage of the extracted bark piece compared with manual peeling | Reported |
| System | Cycle time | Time from scan to completed peel | Reported; compared with manual |
| System | Latency | Time from image capture to joint command | ≤ 2 s |
| Safety | Safety stops | Correct halt on e-stop, joint-limit and low-confidence events | 100% |

### 3.5 Data analysis

Detection results are reported as confusion matrices and the metrics above, overall and per capture condition. Because the number of bench trials is small, proportions such as the peel success rate are reported with 95% Wilson confidence intervals rather than as bare percentages. Positioning error and cycle time are summarised by mean, standard deviation and range. The baseline and the transfer-learning model are compared on the same test images, and the difference is examined using a paired bootstrap over images. Analysis is scripted in Python with NumPy, pandas, scikit-learn and Matplotlib, and the scripts are kept with the code so that every table and figure can be regenerated.

### 3.6 Ethics privacy security and responsible AI

**Ethical clearance.** As stated in the Topic Assessment Form, the project involves no human subjects, personal data or sensitive information, so formal ethics approval was not required. Informal discussions with farmers and agricultural officers during field visits were conducted with verbal consent and no identifying information was recorded. [Add the dates and number of such visits if recorded.]

**Plant material.** Stems were taken with the permission of the owner or from project-managed plants, and no protected material was used.

**Operational safety.** The arm carries a blade and is driven by motors that can pinch or cut. Controls include joint-limit enforcement, speed limits during peeling, a hardware emergency stop that removes servo power, a guarded work area, and a rule that no person enters the work area while the arm is powered. These follow the risk-assessment approach of ISO 12100 [22] and the safety concepts of ISO 10218-1 [23], applied by analogy because the prototype is a research device, not a certified industrial robot.

**Security and privacy.** Communication between the component and the cloud uses HTTPS with token-based authentication, and images are stored only in the project's access-controlled storage. No images of people are captured; if a person appears in a frame during field capture the image is discarded.

**Responsible AI.** The detector is trained on a limited set of sites and conditions, so its performance may not carry over to other cultivars, regions or lighting. The system is therefore designed with a confidence threshold below which it does not act and asks the operator for confirmation, and with a manual-override mode. These behaviours are tested in Chapter 5. Use of generative AI tools in preparing this thesis is disclosed in Appendix B.

### 3.7 Reproducibility and research artefacts

To allow the work to be reproduced, the following artefacts are retained: the annotated dataset with its split lists and labelling guidelines; training and evaluation scripts with fixed random seeds and a requirements file listing library versions; trained model weights and configuration; the calibration files and the procedure used to create them;

the controller firmware and the bill of materials with part specifications; and the raw logs of all bench trials. [Insert repository URL and data-sharing arrangement, subject to the project's intellectual-property position described in Section 8.4.]

# 4 PRODUCT AND SOLUTION DESIGN

## 4.1 Requirements

**Stakeholders.** The system is intended for four groups: cinnamon plantation owners and smallholders, who need less dependence on scarce peelers; agricultural technicians and farm operators, who run the machine and watch the dashboard; system administrators and developers, who maintain models and software; and agricultural researchers, who analyse the collected data. Two personas, a medium-scale estate owner and an agricultural-systems technician, are developed in Section 8.1. Requirements were derived from these needs, from the problem analysis in the project proposal, and from observations during field visits [summarise what was observed and who was consulted].

The functional requirements of the Member 3 component are listed in Table 4.1, and the non-functional requirements in Table 4.2.

**Table 4.1: Functional requirements of the Member 3 component**

| ID | Requirement | Priority | Verified by |
|---|---|---|---|
| FR1 | Capture images of the cinnamon stem at a fixed pose with a 1080p camera | Must | T-01 |
| FR2 | Pre-process images (noise reduction, lighting normalisation, resizing to 512 × 512) | Must | T-02 |
| FR3 | Detect peelable bark regions and return bounding boxes with confidence scores | Must | E1, T-03 |
| FR4 | Extract the bark boundary and a cut line inside the selected region | Must | E1, T-04 |
| FR5 | Convert the cut line to robot-frame coordinates using the stored calibration | Must | E3, T-05 |
| FR6 | Check the harvest-readiness flag from the cloud before acting; allow operator override with explicit confirmation | Must | T-09 |
| FR7 | Compute joint commands by inverse kinematics with joint-limit checks and send them to the controller | Must | T-06, T-07 |
| FR8 | Execute slitting and peeling along the cut line at a limited speed and stop immediately on emergency stop | Must | E4, T-10 |
| FR9 | Inspect the result, grade bark integrity and record the outcome (status, time, image) to the cloud | Should | T-11 |
| FR10 | Provide start, stop, pause, manual/automatic mode and emergency controls through the dashboard | Should | T-12 |

**Table 4.2: Non-functional requirements**

| Attribute | Requirement |
|---|---|
| Performance | Image-to-command latency ≤ 2 s; detection at ≥ 1 frame per second on the target computer |
| Accuracy | End-effector positioning error ≤ 5 mm; repeatability ≤ 2 mm |
| Safety | No motion when the e-stop is active; all commands pass joint-limit and speed checks; arm halts if the heartbeat from the host is lost |
| Reliability | Recover from camera or communication failure without unsafe motion; retry limits before asking the operator |
| Usability | Operator can calibrate, start and stop a cycle by following a one-page procedure; dashboard status is readable at a glance |
| Security | HTTPS and token authentication for all cloud communication |
| Maintainability | Vision, coordinate, control and cloud modules are separate so any one can be replaced without changing the others |
| Cost | Robot mechanical and electronic parts within the LKR 120,000 allowance in the project budget |

## 4.2 Component architecture within the complete solution

Figure 4.1 places the component in the complete platform. The vision service reads frames from the camera, the coordinate service converts detections to robot coordinates, and the robot controller drives the arm. A harvest gate decides whether a cycle may start, using the readiness flag published by Member 2's models through the cloud API, the operator's mode selection and the state of the emergency stop. After peeling, the arm returns to the scan pose so that the camera can inspect the result, and the outcome and images are sent to the cloud for display on the dashboard.

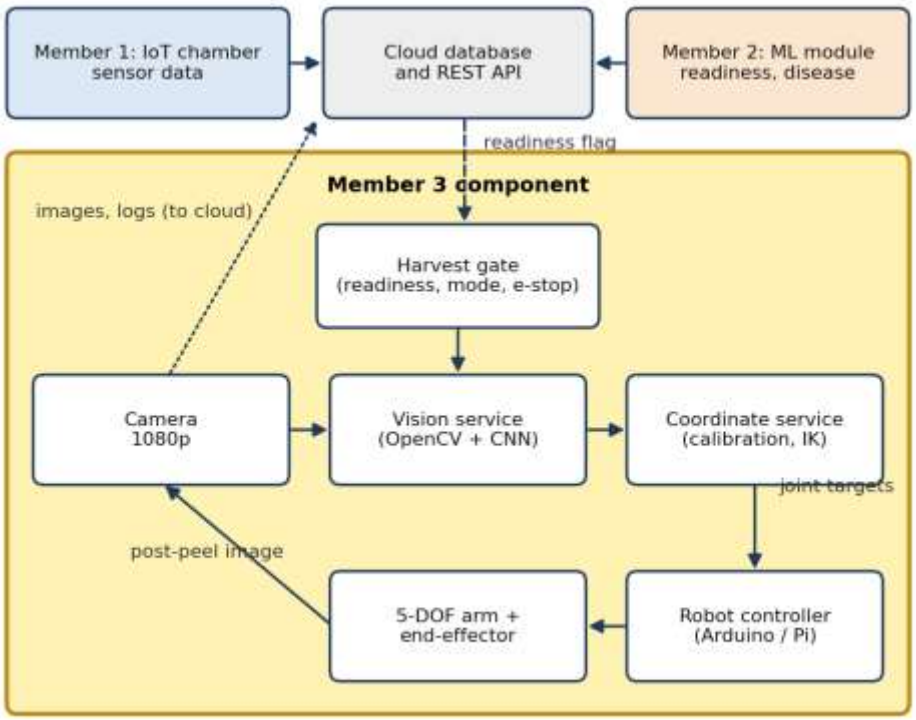



**Figure 4.1: Component architecture within the complete solution**

Table 4.3 lists the interfaces. Each is a narrow, documented data contract so that the three members can develop and test their parts independently.

**Table 4.3: Interfaces of the Member 3 component**

| ID | From → To | Content | Mechanism |
|---|---|---|---|
| I1 | Camera → Vision service | Image frames (1080p) | USB/CSI camera driver |
| I2 | Vision → Coordinate service | Cut line in pixels, box, confidence | In-process call or JSON |
| I3 | Coordinate service → Controller | Joint targets, speed, sequence number | Serial (ASCII frames) |
| I4 | Cloud (Member 2) → Harvest gate | Readiness flag, confidence, predicted date | HTTPS GET, JSON |
| I5 | Component → Cloud | Harvest event record and images | HTTPS POST, JSON/multipart |
| I6 | Dashboard → Component | Start, stop, pause, mode | HTTPS, JSON |

### 4.3 Design alternatives and decisions

The main design decisions, the alternatives considered and the reasons for each choice are summarised in Table 4.4. The dominant constraints were cost, the need to work with a small self-collected dataset, and the limited accuracy of hobby servos.

**Table 4.4: Design alternatives and decisions**

| Decision | Alternatives considered | Choice and rationale |
|---|---|---|
| Vision model | (a) Image-level classification CNN; (b) one-stage detector; (c) semantic segmentation (e.g. U-Net [24]) | (b) with classical boundary refinement. A detector gives locations, not just a yes/no answer, and needs only box labels. Segmentation would give better boundaries but needs costly pixel-level labels, so it is left for future work. (a) is kept as the baseline. |
| Depth / 3D position | (a) Monocular camera with plane homography; (b) depth camera; (c) stereo pair | (a). The stem lies on a known fixture plane, so a homography is sufficient and avoids extra cost. Stem height is taken from the fixture. |
| Where inference runs | (a) Cloud; (b) local workstation or Raspberry Pi; (c) microcontroller | (b). Local inference keeps latency low and works without connectivity; the cloud is used for storage, dashboard and readiness data. [Confirm the computer actually used.] |
| Actuators | (a) MG996R hobby servos; (b) stepper motors; (c) smart serial servos | (a), within the budget and easy to drive; accuracy limits are measured in E3 and discussed in Section 7.5. |
| Control interface | (a) Direct serial protocol to Arduino; (b) ROS [25] | (a) for simplicity and transparency. ROS is considered for later versions. [Update if ROS was used.] |
| End-effector | (a) Slitting blade plus lifting/peeling jaw; (b) gripper only | (a). Peeling requires a controlled slit with a depth stop before the bark is lifted. [Confirm final design and add photographs in Chapter 5.] |

### 4.4 Detailed design

The detailed design follows the processing pipeline in Figure 4.2. An image is captured and pre-processed, the detector finds the peelable region, boundary extraction produces a cut line, the cut line is mapped into the robot frame, inverse kinematics converts it to joint targets, and a smoothed trajectory is executed under a safety monitor. The sections below describe each stage.

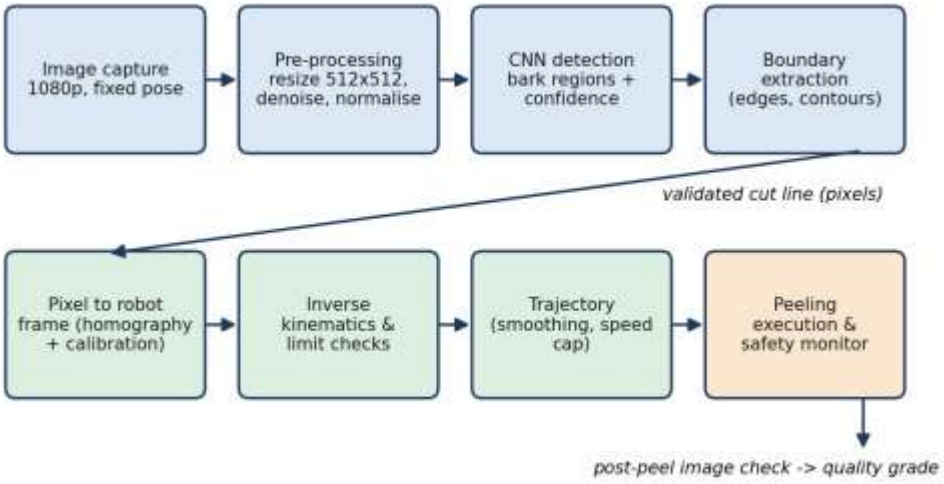



**Figure 4.2: Processing pipeline from image capture to peeling**

**Image acquisition and pre-processing.**

The camera is fixed above the work plane so that every capture has the same geometry as during calibration. Each frame is converted to a working copy at 512 × 512 pixels, denoised with a Gaussian or bilateral filter, and brightness-normalised (for example by contrast-limited adaptive histogram equalisation on the luminance channel) to reduce the effect of sunlight and shade [6]. The original frame is retained for logging.

**Bark detection.**

Two models are built. The **baseline** is the small CNN in Table 4.5, which classifies an image or patch as containing a peelable region or not. The **main model** is a one-stage detector with a lightweight backbone, initialised from pre-trained weights and fine-tuned on the cinnamon dataset [2], [3], [4], [26]. It outputs bounding boxes for the two classes with confidence scores; boxes below a confidence threshold (initially 0.5, tuned on the validation set) are discarded and overlapping boxes are merged by non-maximum suppression.

**Table 4.5: Baseline CNN architecture**

| Layer | Description | Output size |
|---|---|---|
| Input | RGB image or patch | 512 × 512 × 3 |
| Convolution 1 | 32 filters, stride 2, ReLU | 256 × 256 × 32 |
| Max pooling | 2 × 2 down-sampling | 128 × 128 × 32 |
| Convolution 2 | 64 filters, stride 2, ReLU | 64 × 64 × 64 |

| Layer | Description | Output size |
|---|---|---|
| Fully connected | Flatten followed by dense layer, ReLU, dropout | 128 |
| Output | Sigmoid: peelable region present / absent | 1 |

**Boundary extraction and cut line.**
The detector gives a box, not an exact edge. Within the highest-confidence peelable box the image is converted to HSV (or Lab) space, smoothed, and passed through Canny edge detection [8] followed by morphological closing. The largest connected contour is taken as the bark region. Principal component analysis of the contour points gives the stem's long axis, and the **cut line** is the line along this axis inside the region, offset from the edges by a safety margin. The line is accepted only if its length, straightness and position inside the box meet thresholds; otherwise the cycle is rescanned.

**Pixel-to-robot coordinate mapping.**
The camera's intrinsic parameters and distortion are estimated from a printed chessboard [9]. Pixels are then mapped to metric positions on the work plane by a homography estimated from at least four reference markers whose plane coordinates are known [10]:

$$s \cdot [u, v, 1]^T = H \cdot [X, Y, 1]^T \qquad P_{robot} = R \cdot [X, Y, Z_{plane}]^T + t$$

Here (u, v) are pixel coordinates, (X, Y) are plane coordinates, $Z_{plane}$ is the known fixture height, and R and t are the rotation and translation from the plane frame to the robot base frame, estimated by touching the reference markers with the end-effector tip. Sampling the cut line every few millimetres yields the path in the robot frame.

**Arm kinematics and servo mapping.**
The arm has five degrees of freedom (Table 4.6): base rotation, shoulder, elbow, wrist pitch and wrist roll. After the base angle is set from the target's bearing, the shoulder and elbow are solved in the vertical plane using the law of cosines [11], [12]:

$$\theta_1 = \operatorname{atan2}(y, x) \qquad \cos\theta_3 = (r'^2 + z'^2 - L_1^2 - L_2^2) / (2 L_1 L_2) \qquad \theta_2 = \operatorname{atan2}(z', r') - \operatorname{atan2}(L_2 \sin\theta_3, L_1 + L_2 \cos\theta_3)$$

where $r'$ and $z'$ are the horizontal and vertical distances to the wrist after subtracting the tool length, and $L_1$ and $L_2$ are the link lengths. Solutions outside the joint limits or

workspace are rejected. Joint angles are converted to PWM pulse widths using a per-joint calibration table rather than the nominal 500–2500 µs range, because individual servos differ.

**Table 4.6: Robot and camera parameters**

| Parameter | Value |
|---|---|
| Motor type | MG996R servo motor (stall torque about 9.4–11 kg·cm depending on supply voltage) |
| Degrees of freedom | 5 |
| Control method | AI-generated Cartesian (X-Y-Z) targets converted by inverse kinematics |
| Free-travel speed | Up to 0.5 m/s nominal at the end-effector [verify achievable value] |
| Speed during contact | Limited to [__ mm/s] during slitting and peeling |
| Camera | 1080p, fixed pose above work plane |
| Link lengths $L_1$, $L_2$, tool length | [measure and insert] |
| Joint limits | [insert per joint] |

**Motion and peeling strategy.**
Motion is planned in Cartesian space: the end-effector moves above the start of the cut line, descends to a depth stop, follows the path with linear interpolation in small steps under a trapezoidal speed profile, lifts the bark flap and performs the peeling pull. Contact speed is deliberately far lower than free-travel speed. After peeling the arm retracts and the camera inspects the result. Bark integrity is graded as A (intact piece of at least 90% of the planned length), B (60–90%) or C (shorter or torn). [Adjust grading thresholds to the grading practice found in field visits.]

**Control logic.**
Operation is organised as the state machine in Figure 4.3. A cycle starts in IDLE and moves to SCAN when the operator starts it. In VERIFY the harvest gate checks the readiness flag and the model confidence; if the stem is not ready the machine waits, and if confidence is low it rescans up to a set number of times before asking the operator. PLAN computes and validates the path, APPROACH, PEEL and INSPECT execute and check it, and DONE logs the outcome. Any state can transition to

FAULT/E-STOP, from which a cycle can be resumed only after a deliberate operator reset.

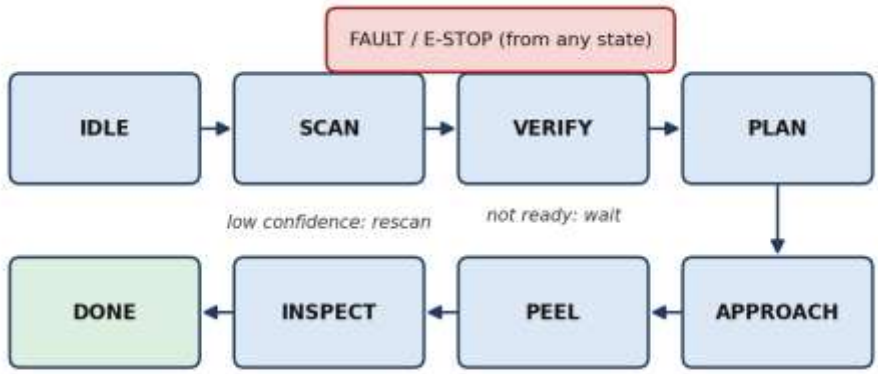



**Figure 4.3: State machine of the harvest cycle**

**Safety design.**
Safety is built into both software and hardware. The host sends a heartbeat every 200 ms and the controller stops the arm if it is not received within 500 ms. The controller itself enforces joint limits and maximum step size, independent of the host. A hardware emergency-stop switch cuts servo power. The work area is guarded and the blade is retracted whenever the arm is not in the PEEL state.

### 4.5 Standards and professional practice

Risk assessment followed the iterative approach of ISO 12100 [22], and the safeguards above reflect the concepts of ISO 10218-1 for robots [23]; the prototype is not claimed to be certified under either standard. [Confirm the edition of ISO 10218-1 used.] Software follows PEP 8 style, is kept under Git version control, and is documented so that modules can be replaced. References follow the IEEE style. Open-source components are used under their licences, as discussed in Section 8.4. The work observes the IEEE Code of Ethics with respect to safety, honesty about limitations, and acknowledgement of the work of others.

# 5 IMPLEMENTATION INTEGRATION AND TESTING

## 5.1 Development environment and technology stack

**Table 5.1: Development environment and technology stack**

| Layer | Technology | Purpose |
|---|---|---|
| Language | Python 3.x [version] | Vision, coordinate service, backend |
| Deep learning | TensorFlow/Keras [27]; [detector library, e.g. Ultralytics YOLO [26]] | Baseline CNN; detector training and inference |
| Computer vision | OpenCV [6] | Pre-processing, edges, contours, calibration, homography |
| Annotation | LabelImg / Roboflow | Bounding-box labelling |
| Numerics and analysis | NumPy, pandas, scikit-learn, Matplotlib | Kinematics, metrics, plots |
| Backend and dashboard | Flask (REST API); React (web dashboard) | Interface to cloud and operator |
| Storage | [MongoDB / Firebase / MySQL]; [AWS S3 / Google Cloud Storage] | Events, images, logs |
| Controller | Arduino / Raspberry Pi with servo driver | PWM generation, safety checks |
| Hardware | 5-DOF arm, MG996R servos, 1080p camera, regulated supply, e-stop | Prototype robot |
| Training machine | Intel i7, 16 GB RAM, CUDA-capable GPU, 512 GB SSD [confirm] | Model training |

## 5.2 Component implementation

**Data pipeline.**
Images were captured under the conditions listed in Table 3.1, copied to a structured directory, annotated, and checked for label errors. A script created the stem-level split lists and applied augmentation to the training subset only. [State final counts and any images removed, with reasons.]

**Model training.**
The baseline CNN and the detector were trained with the settings in Table 5.2. Training used the Adam optimiser [28], early stopping on validation loss and a fixed random seed. Hyper-parameters were tuned on the validation set only; the test set was used once for each final model.

**Table 5.2: Training configuration**

| Setting | Baseline CNN | Detector |
|---|---|---|
| Initialisation | Random | Pre-trained weights [name] |
| Input size | 512 × 512 | 512 × 512 |
| Optimiser / learning rate | Adam / [1e-3] | [optimiser / value] |
| Batch size | [16] | [16] |
| Epochs | [up to 100, early stopping] | [__] |
| Confidence / NMS thresholds | Decision threshold [0.5] | [0.5] / [0.45] |
| Training time and hardware | [__] | [__] |

**Calibration.**
Calibration is a one-off procedure repeated whenever the camera or fixture is moved: (1) capture [N] chessboard images and compute intrinsics and distortion; (2) place reference markers on the work plane and compute the homography; (3) touch each marker with the end-effector to estimate R and t; (4) measure each servo's pulse-to-angle table; (5) store all parameters in a versioned calibration file. The procedure and resulting values are given in Appendix A.

**Controller firmware and protocol.**
The host sends ASCII frames of the form M,seq,j1,j2,j3,j4,j5,speed,checksum, and the controller replies A,seq when the frame is valid and has been accepted or E,code when it is rejected. Frames with a bad checksum, a joint outside its limits or a step above the maximum are refused. A heartbeat frame is exchanged every 200 ms, and the controller halts if none arrives for 500 ms. [Confirm the frame format matches the final firmware.]

**Cycle control.**
The host application implements the state machine of Figure 4.3. The core of one cycle is summarised below.

```
def run_cycle(cam, model, calib, robot, cloud, mode):
    if mode == 'AUTO' and not cloud.get_readiness().ready:
        return 'WAIT'                       # harvest gate
    for attempt in range(MAX_RESCANS):
        frame = cam.capture()
        det = model.detect(preprocess(frame))
        line = extract_cut_line(frame, det)
        if det.conf >= CONF_MIN and line.valid:
            break
```

```
    else:
        return ask_operator()                # low confidence
    path = calib.pixels_to_robot(line.points)
    joints = [ik(p) for p in path]           # reject if out of limits
    robot.move_to_start(joints[0])
    robot.follow(joints, speed=PEEL_SPEED)   # heartbeat + e-stop
monitored
    robot.peel()
    grade = inspect(cam.capture())
    cloud.post_event(grade, frame)
    return grade
```

**Mechanical assembly.**
[Describe the construction of the arm, mounting, fixture and end-effector, and insert photographs and key dimensions as figures. State any parts that were 3D-printed or fabricated.]

### 5.3 Integration with the other components

Integration with Member 1 and Member 2 takes place through the cloud API, so that the robot does not depend on the internal design of either component. Table 5.3 lists the endpoints used. [Align endpoint names and fields with the implemented backend.]

**Table 5.3: Cloud API endpoints used by the Member 3 component**

| Endpoint | Method | Purpose |
|---|---|---|
| /api/readiness/{batch_id} | GET | Obtain the harvest-readiness flag, confidence and predicted date produced by Member 2's models |
| /api/harvest/events | POST | Upload the outcome of a cycle: time, grade, confidence, images |
| /api/harvest/command | GET / POST | Read or set start, stop, pause and mode from the dashboard |
| /api/harvest/status | GET | Current state of the cycle for display |

The dashboard shows the live camera view with the detected region, the planned cut line, the current state and the log of completed cycles. Preliminary mock-ups produced during proposal development guided the layout. [Insert screenshots of the final dashboard as figures.] Integration tests with Members 1 and 2 are reported in Section 6.4.

### 5.4 Verification and software testing

Testing was organised at unit, integration, hardware and safety levels, as listed in Table 5.4. Software tests were written with pytest and run automatically; hardware and safety tests were performed manually following written checklists. [Fill in the result column from the actual test log.]

**Table 5.4: Test plan and results**

| ID | Test | Expected result | Result |
|---|---|---|---|
| T-01 | Camera capture at fixed pose | 1080p frame saved with timestamp | [Pass/Fail] |
| T-02 | Pre-processing on dark, bright and blurred images | Output 512 × 512, no crash, histogram within range | [ ] |
| T-03 | Detector on test images, empty scene, no stem | Boxes on stems; no boxes on empty scene | [ ] |
| T-04 | Cut-line extraction on valid and invalid boxes | Valid line accepted; invalid rejected | [ ] |
| T-05 | Homography on known reference points | Error within tolerance | [ ] |
| T-06 | Inverse kinematics for reachable points | Joint angles reproduce target by forward kinematics | [ ] |
| T-07 | Inverse kinematics for unreachable points | Rejected; no command sent | [ ] |
| T-08 | Serial protocol with corrupted frames | Frame refused; error returned | [ ] |
| T-09 | Harvest gate: not ready / ready / override | No motion / motion / motion only after confirmation | [ ] |
| T-10 | Emergency stop during motion | Servo power removed; arm halts | [ ] |
| T-11 | Event upload with network available / unavailable | Uploaded; or queued and retried | [ ] |
| T-12 | Dashboard start, pause, stop | State changes as commanded | [ ] |
| T-13 | Heartbeat loss | Controller halts within 500 ms | [ ] |

### 5.5 Deployment and operation

The prototype is deployed as a bench installation. The camera and fixture are mounted on a rigid frame; the host computer runs the vision and coordinate services and the Flask backend; the controller is connected by USB serial; and the dashboard is reached through a web browser. The operating procedure is: power up with the e-stop engaged; check the calibration file and run the quick calibration check on the reference markers;

load a stem into the fixture; select automatic or manual mode; start the cycle; observe the first moves with a hand near the e-stop; and review the logged result. Maintenance consists of checking servo horns and fasteners, cleaning the camera lens, sharpening or replacing the blade, and re-running calibration after any mechanical change.

# 6 RESULTS AND VALIDATION

[IMPORTANT: every cell and blank in this chapter must be replaced with measured results from your own experiments before submission. No result values are supplied by this draft. Delete this note afterwards.]

## 6.1 Research results

This section reports experiments E1 and E2 (RQ1). The baseline CNN and the detector were evaluated once on the [300]-image test set. At image level, an image counts as positive for the detector if at least one peelable box exceeds the confidence threshold. Table 6.1 compares the two models.

**Table 6.1: Test-set results for the baseline CNN and the detector**

| Model / level | Precision | Recall | F1 | mAP@0.5 |
|---|---|---|---|---|
| Baseline CNN (image level) | [ ] | [ ] | [ ] | n/a |
| Detector (image level) | [ ] | [ ] | [ ] | n/a |
| Detector (box level, IoU ≥ 0.5) | [ ] | [ ] | [ ] | [ ] |

The detector achieved a precision of [___] and a recall of [___] at image level, compared with [___] and [___] for the baseline. At box level the detector reached mAP@0.5 of [___]. The extracted bark region had a mean boundary IoU of [___] against the annotated regions. [Insert the confusion matrix, precision–recall curve and examples of correct and incorrect detections as figures, and cite them here.]

Experiment E2 examined robustness to capture conditions. Table 6.2 reports the results by condition.

**Table 6.2: Detector performance by capture condition**

| Condition | Images | Precision | Recall |
|---|---|---|---|
| Sunny daylight | [ ] | [ ] | [ ] |
| Overcast | [ ] | [ ] | [ ] |
| Shade | [ ] | [ ] | [ ] |
| Indoor white light | [ ] | [ ] | [ ] |
| Plain background | [ ] | [ ] | [ ] |
| Cluttered background | [ ] | [ ] | [ ] |

Performance was lowest in [condition], where [describe typical errors, e.g. glare on the bark, shadows resembling edges, or leaves overlapping the stem].

### 6.2 Product and technical results

**Positioning accuracy (E3).** Table 6.3 reports the difference between commanded and achieved end-effector positions.

**Table 6.3: Positioning accuracy and repeatability**

| Measure | Result | Target |
|---|---|---|
| Target points tested × repetitions | [N × k] | — |
| Mean positioning error (mm) | [ ] | ≤ 5 |
| Standard deviation / maximum (mm) | [ ] | — |
| Trials within 5 mm (%) | [ ] | ≥ 90 |
| Repeatability, standard deviation (mm) | [ ] | ≤ 2 |

**Integrated bench trials (E4).** Table 6.4 summarises complete harvest cycles on [N] stems.

**Table 6.4: Integrated bench trial results**

| Measure | Automated | Manual (if measured) |
|---|---|---|
| Stems tested | [ ] | [ ] |
| Successful extractions (95% CI) | [n / N (CI)] | [ ] |
| Grade A / B / C | [/ /] | [/ /] |
| Mean cycle time (s) | [ ] | [ ] |
| Operator interventions | [ ] | — |
| Main failure modes | [e.g. blade slipping off line, bark tearing, low confidence] | — |

**Latency (E5).** Table 6.5 gives the time spent in each stage.

**Table 6.5: Pipeline latency by stage**

| Stage | Mean (ms) | Maximum (ms) |
|---|---|---|
| Capture and pre-processing | [ ] | [ ] |
| Detection | [ ] | [ ] |

| Stage | Mean (ms) | Maximum (ms) |
|---|---|---|
| Boundary and cut line | [ ] | [ ] |
| Coordinate mapping and IK | [ ] | [ ] |
| Total image-to-command | [ ] | [ ] (target ≤ 2000) |

### 6.3 User or stakeholder validation

No formal user study with human participants was performed, in line with the ethics statement in Section 3.6. The prototype was demonstrated to [supervisors / agricultural officers / estate representatives, with dates], and their comments were recorded informally with verbal consent and without personal identifiers. Table 6.6 summarises the feedback and the resulting actions.

**Table 6.6: Informal stakeholder feedback**

| Stakeholder group | Feedback | Action taken |
|---|---|---|
| [ ] | [ ] | [ ] |
| [ ] | [ ] | [ ] |

[If no stakeholder demonstration took place, state this plainly and list it under limitations in Section 7.5 instead of leaving the table empty.]

### 6.4 Integration results

Integration with Member 1's sensing/cloud layer and Member 2's readiness model was tested with the scenarios in Table 6.7.

**Table 6.7: Integration scenarios**

| Scenario | Expected behaviour | Observed |
|---|---|---|
| Readiness flag false | Robot stays in VERIFY and does not move | [ ] |
| Readiness flag true | Cycle proceeds to PLAN and PEEL | [ ] |
| Operator override with confirmation | Cycle proceeds and the override is logged | [ ] |
| Cloud unreachable | Cycle result queued locally and uploaded later | [ ] |
| Dashboard stop during PEEL | Arm halts safely and state is shown | [ ] |

## 6.5 Summary of findings

**Table 6.8: Summary of findings against research questions**

| Question | Finding | Evidence |
|---|---|---|
| RQ1: detection accuracy | [Summarise] | Table 6.1, Table 6.2 |
| RQ2: coordinate accuracy | [Summarise] | Table 6.3 |
| RQ3: integrated performance | [Summarise] | Table 6.4, Table 6.5, Table 6.7 |

Hypotheses H1 to H3 were [supported / partly supported / not supported] because [reason].

# 7 DISCUSSION

## 7.1 Interpretation of findings

[Write this section after the results are final.] The discussion should address four points. First, whether the detector improved on the baseline and why: a detector provides location as well as presence, benefits from pre-trained features when only a small dataset is available [3], and so is expected to outperform a small CNN trained from scratch; the measured difference of [___] should be interpreted against this expectation. Second, which conditions caused errors and whether these match the known sensitivity of vision methods to lighting [15]. Third, how much of the end-to-end error is attributable to vision, to calibration and to the servos, using the geometry-level results to separate them. Fourth, which failure modes occurred in the bench trials and what they imply for the end-effector and control design.

## 7.2 Comparison with literature baselines and existing solutions

Direct numerical comparison with fruit-harvesting studies is not appropriate, because the targets, datasets and metrics differ [16], [17], [18], [19]. Qualitatively, those systems detect compact objects and detach them, whereas this prototype delineates a region on a continuous surface and carries out a cut-and-peel motion. Compared with manual peeling, the prototype is expected to offer consistency and reduced physical strain but, as a bench-scale device, not yet the throughput of an experienced worker; the measured cycle times in Table 6.4 should be used to state this precisely. [Insert the numbers reported by [16] and [17] for their detectors, with the caveat above.]

## 7.3 Research contribution

- A labelled image dataset of cinnamon stems and bark for harvest-region detection under varied lighting [state whether it will be shared].
- Evidence on whether a transfer-learning detector combined with classical boundary extraction can delineate peelable bark regions, compared with a baseline CNN.
- A low-cost calibration method that maps a single camera's detections to a servo-driven arm through a plane homography, with measured accuracy.

- A design in which a prediction-based readiness signal gates a vision-guided robot, demonstrated end to end with the other project components.

### 7.4 Product contribution

The product contribution is a working bench-scale prototype consisting of the vision and coordinate software, the controller firmware and safety logic, the mechanical arm and end-effector, and the dashboard interface. It demonstrates the complete cycle from readiness check to peeled bark and logged result, and provides a base on which later versions can be built.

### 7.5 Limitations and threats to validity

- **External validity.** The dataset covers one species, limited sites and growth stages; performance on other cultivars, estates or seasons is unknown. Peeling was tested on cut stems in a fixture, not on standing plants in the field.
- **Internal validity.** Annotation is subjective; inter-annotator agreement was [ ___ ]. Small test sets give wide confidence intervals, so differences between models may not be statistically reliable. Lighting during bench trials may differ from field lighting.
- **Construct validity.** Bounding-box detection approximates, but is not identical to, the expert's judgement of peel readiness. The integrity grade is a simple proxy for quill quality and has not been validated against buyers' grading standards.
- **Hardware.** MG996R servos have backlash, limited torque and temperature-dependent behaviour; this restricts accuracy and the force that can be applied. The monocular set-up cannot detect variations in stem height or curvature outside the fixture plane.
- **Dependence on other components.** The readiness signal is produced by Member 2's models and was treated as given; its errors are not evaluated here.
- **Safety and certification.** The prototype is a research device and has not been assessed against machinery-safety standards for commercial use.

### 7.6 Implications

If the prototype's performance is confirmed in larger trials, vision-guided mechanisation could reduce dependence on scarce skilled peelers, lower injury risk, and make bark quality more uniform. For researchers, the dataset and pipeline provide a starting point for work on bark-based harvesting, which is largely absent from the agricultural-robotics literature [20], [21]. For adoption, the findings emphasise that affordability, clear safety practice and acceptance by peeling communities will matter as much as technical accuracy.

# 8 COMMERCIALISATION AND SUSTAINABILITY

## 8.1 Value proposition and target users

The system targets cinnamon plantation owners, small and medium-scale growers, agricultural cooperatives, smart-agriculture technology providers and agricultural research organisations. Two personas from the project proposal illustrate the users. **Mr. Nimal Perera** manages a medium-scale estate in the Southern Province. His goals are to improve harvesting efficiency, reduce dependence on skilled manual labour and maintain consistent bark quality; his challenges are a shortage of skilled peelers, high labour costs during the harvest season and inconsistent quality. **Ms. Sanduni Jayasinghe** is an agricultural-systems technician at a smart-agriculture company who monitors environmental conditions, deploys harvesting technology and wants to improve operational efficiency through an IoT dashboard and AI-assisted bark detection.

The value proposition has four parts: automation of bark detection and peeling, relief from labour shortages, more consistent bark quality, and data-driven decisions through the integrated sensing and prediction platform.

## 8.2 Market and competitive position

Cinnamon is an export-oriented crop in which Sri Lanka holds a leading position in true cinnamon [CITE market data]. Competing alternatives are manual peeling, which is flexible but slow and variable, and general agricultural robots, which are designed for other crops and cannot peel bark. No commercial cinnamon-peeling robot was identified in the review (Section 2.4). The competitive advantage of this work is therefore its crop-specific design and its integration of vision, prediction and robotic control in one platform. Its weaknesses are low maturity and the unproven throughput of a hobby-grade prototype.

## 8.3 Business or sustainability model

Three revenue models are considered: (i) **hardware and software sales**, where growers buy the machine with its software; (ii) **subscription**, where users pay monthly or annually for cloud monitoring, analytics and updates; and (iii) **technology**

**licensing**, where equipment makers integrate the vision module into their own machines. A tiered offering could be: Basic (AI harvesting module and basic dashboard), Standard (adds environmental monitoring and predictive harvest alerts) and Premium (full robotic automation, advanced analytics and integration with other smart-farming platforms).

Table 8.1 shows the development budget set out in the project proposal for this component. The proposal budget does not list the controller, power supply, fixture, blade or enclosure, nor field-trial costs, which would be needed in a commercial cost estimate. [Add these items and estimate a unit production cost and break-even volume.]

**Table 8.1: Development budget for the Member 3 component (from the project proposal)**

| Item | Estimated cost (LKR) |
|---|---|
| AI development workstation (i7, 16 GB RAM, GPU, 512 GB SSD) | 250,000 |
| Camera module (1080p) | 20,000 |
| Robotic machine components (servo motors, arm parts) | 120,000 |
| Development tools and licences | 20,000 |
| AI libraries (TensorFlow, OpenCV, Keras) | Free / open source |
| Cloud storage and compute | 50,000 |
| Research and development effort (200 h × LKR 1,500) | 300,000 |
| **Total** | **760,000** |

### 8.4 Scalability intellectual property and compliance

**Scalability.** The architecture is modular and cloud-backed, so multiple machines and sites can report to one platform, and models can be retrained as more data are collected. Scaling the hardware would require a sturdier arm with higher-precision actuators and a mechanism for presenting stems automatically.

**Intellectual property.** The software code and algorithms are protected by copyright; the end-effector and peeling mechanism may be considered for patent protection; the system name and branding by trademark; and commercial partnerships by licensing agreements. The underlying frameworks are open source, so the novelty lies in the

integration and the mechanism rather than in the libraries. The dataset is a valuable asset and its release should be decided with the supervisors.

**Licence compliance.** TensorFlow and OpenCV are released under permissive licences, but some popular detector implementations, including Ultralytics YOLO, are distributed under the AGPL-3.0 licence with a separate commercial licence [verify current licence terms]. A commercial product would need either a commercial licence or an alternative detector with a compatible licence.

**Compliance.** A commercial machine would need to meet applicable machinery-safety requirements [CITE the Sri Lankan and export-market requirements]. If farm records linked to identifiable individuals were stored in the cloud, Sri Lanka's Personal Data Protection Act, No. 9 of 2022, may apply [confirm]. The research data in this thesis contain no personal data.

**Social responsibility.** Mechanisation must not simply remove livelihoods. The system is positioned to assist skilled workers and to take over the most hazardous steps, with training and phased introduction recommended.

## 8.5 Contribution to the Sustainable Development Goals

**Table 8.2: Contribution to the Sustainable Development Goals**

| SDG | Contribution |
|---|---|
| SDG 2: Zero Hunger | Supports productive, resilient agriculture through precision monitoring and reduced harvest losses |
| SDG 3: Good Health and Well-being | Reduces exposure of workers to blade injuries and repetitive strain by automating peeling |
| SDG 8: Decent Work and Economic Growth | Automates labour-intensive work and supports stable rural incomes, while recognising the need for a just transition for workers |
| SDG 9: Industry, Innovation and Infrastructure | Introduces AI, IoT and robotics into traditional agriculture |
| SDG 12: Responsible Consumption and Production | Enables more uniform, higher-quality output with less waste and better use of resources |

# 9 CONCLUSION AND FUTURE DIRECTIONS

## 9.1 Conclusion

This thesis addressed the manual, skill-dependent nature of cinnamon bark harvesting by designing and building an AI-guided robotic machine that detects peelable bark regions from camera images and guides a five-degree-of-freedom arm to extract the bark. The work combined a CNN-based detector with classical boundary extraction, a calibration-based mapping from pixels to robot coordinates, inverse-kinematics motion control with layered safety measures, and integration with the readiness predictions and dashboard of the wider project.

[Summarise the key quantitative findings: detection precision/recall, positioning error, peel success rate and cycle time, and state whether the targets in Table 3.2 were met.] The results show that [a low-cost, vision-guided approach is / is not yet sufficient for reliable bark extraction], and they identify [the principal limiting factors].

## 9.2 Achievement of objectives

**Table 9.1: Achievement of the research objectives**

| Objective | Evidence | Status |
|---|---|---|
| RO1: bark-detection pipeline | Sections 4.4, 5.2; Table 6.1, Table 6.2 | [Met / partly / not met] |
| RO2: deep-learning model and baseline comparison | Table 6.1 | [ ] |
| RO3: accurate harvesting coordinates | Table 6.3 | [ ] |
| RO4: integration with robot control and Member 2 | Sections 5.3, 6.4 | [ ] |
| RO5: performance evaluation | Table 6.4, Table 6.5 | [ ] |

## 9.3 Individual research and product contributions

The author's research contributions are the dataset, the comparison of a baseline CNN with a transfer-learning detector for bark detection, and the evaluation of a single-camera calibration approach for a low-cost arm. The product contributions are the vision and coordinate software, the controller firmware and safety logic, the integrated bench prototype and its dashboard interface, as described in Sections 1.8 and 7.4.

### 9.4 Future directions

- Collect a larger, multi-site, multi-season dataset covering more cultivars and growth stages, and add pixel-level labels to train a segmentation model for more precise boundaries.
- Add a depth camera or stereo pair so that stems of varying height and curvature can be handled without a fixed plane.
- Introduce force or current sensing in the end-effector for closed-loop control of slit depth and peeling force.
- Replace hobby servos with higher-precision actuators and evaluate ROS-based motion planning.
- Extend the platform to mobile operation in the plantation, including cutting and stem handling.
- Explore multispectral cues from the wider project to improve the judgement of peel readiness.
- Run field trials with estates and compare cost, throughput and bark grade with manual peeling.

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# GLOSSARY

**Bark readiness** – The condition in which the bark of a cinnamon stem separates cleanly from the wood and can be peeled.

**Cut line** – The line along a stem, inside the detected bark region, along which the blade slits the bark.

**Degrees of freedom (DOF)** – The number of independent joint movements of a robotic arm.

**End-effector** – The tool at the end of the arm, here the slitting and peeling tool.

**Harvest gate** – The software check that permits a harvest cycle to start only when readiness, mode and safety conditions are satisfied.

**Heartbeat** – A short message sent regularly between host and controller; its absence causes the arm to stop.

**Homography** – A projective transformation that maps points on one plane to points on another, used here to map image pixels to the work plane.

**Intersection over union (IoU)** – The overlap between a predicted and a true region divided by their combined area.

**Inverse kinematics** – Calculation of the joint angles that place the end-effector at a desired position.

**Mean average precision (mAP)** – The mean over classes of the area under the precision–recall curve of a detector.

**Quill** – A rolled piece of dried cinnamon bark, the standard traded form of true cinnamon.

**Transfer learning** – Starting training from a model pre-trained on a large dataset and fine-tuning it on a smaller task-specific dataset.

# APPENDICES

## Appendix A Research instruments or extended results

**A.1 Labelling guidelines.** [Insert the written annotation rules: what counts as a peelable region, how to treat partially covered bark, minimum box size, handling of knots and branch scars, and examples.]

**A.2 Calibration data.** [Insert the camera intrinsic matrix, distortion coefficients, homography, the R and t estimates and the per-joint pulse-to-angle tables.]

**Table A.1: Calibration record**

| Item | Value | Date |
|---|---|---|
| Camera intrinsics and distortion | [ ] | [ ] |
| Homography H | [ ] | [ ] |
| Plane-to-robot R, t | [ ] | [ ] |
| Servo pulse-to-angle tables | [ ] | [ ] |

**A.3 Extended results and trial log.** [Insert the full confusion matrices, per-trial bench results and any additional plots.]

## Appendix B Academic integrity and AI use disclosure

In accordance with the module policy, the use of generative AI and automated tools is disclosed in Table B.1. The author remains responsible for the accuracy, originality, citations, code, data, analyses and conclusions of this thesis. [The author must verify and complete this table so that it reflects what was actually done.]

**Table B.1: Disclosure of AI and automated tool use**

| Tool | Purpose | Verification performed | Sections affected |
|---|---|---|---|
| Claude (Anthropic) | Drafting the thesis structure and text from the project proposal, topic assessment form, template and a sample thesis; generating diagrams with Matplotlib code | [Author checked technical statements against own work; replaced all placeholders with own data; checked each reference] | All chapters; Figures 1.1, 3.1, 4.1 to 4.3 |
| AI image-generation tools | Conceptual architecture illustrations and user-interface mock-ups used in the project proposal | Marked as conceptual in the proposal | Proposal only [state if reused here] |

| Tool | Purpose | Verification performed | Sections affected |
|---|---|---|---|
| [Other tools, e.g. grammar checkers] | [ ] | [ ] | [ ] |